SSC MTS 2024 question paper with answers

7 October 2024 · Shift 1

82 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 8 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 17 questions

Q.1

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 9% per year rate of interest for 6 years.

(A) 1290
(B) 1350
(C) 1310
(D) 1390

Q.2

Raju sold 4 laptops at a rate of ₹1,54,005 all together and incurred a loss of 25%. At what rate should he sell the laptops to gain a profit of 25%?

(A) ₹2,31,007.5
(B) ₹2,05,340
(C) ₹2,56,675
(D) ₹2,25,820

Q.4

A vendor wants to gain 25% by adding water to milk and selling the mixture at the cost price. Find the ratio of water and milk in the mixture.

(A) 1 : 5
(B) 2 : 7
(C) 1 : 4
(D) 2 : 5

Q.5

A sum of ₹2,380 is lent out into two parts, one at 25% p.a. and another at 20% p.a. . If the total annual income is ₹500, find the money (in ₹) lent at 25% rate.

(A) 320
(B) 560
(C) 480
(D) 240

Q.6

The highest common factor of 84 and 108 is:

(A) 6
(B) 12
(C) 8
(D) 16

Q.7

What is the area of the remaining portion of a circle of radius 14 cm when a square with maximum possible area inscribed in it is removed from it? (Use π= 22/7)

(A) 518 cm^2
(B) 520 cm^2
(C) 616 cm^2
(D) 224 cm^2

Q.8

The speed of a boat in still water is 15 km/h and the speed of the current is 3 km/h. How much distance (in km) can be covered going downstream in 12 minutes?

(A) 3.6
(B) 3.7
(C) 3.5
(D) 3.4

Q.9

A person travels equal distances with speeds of 3 km/hr, 4 km/hr and 8 km/hr, respectively, and takes a total of 425 minutes. What is the total distance (in km) travelled?

(A) 30
(B) 27
(C) 28
(D) 32

Q.10

The marked price of a folding bed is ₹18,000. Aman bought it after two successive discount of 15% and 10%. He spent ₹830 on transportation and sold it for ₹18,440. Find his profit.

(A) ₹3,539
(B) ₹2,869
(C) ₹3,148
(D) ₹3,840

Q.12

A hemispherical bowl is 264 cm round at the brim. Suppose that the bowl is half-full of beverage. If the beverage is served in hemispherical glasses 6 cm in the diameter at the top, how many people would be served?

(A) 1470
(B) 1170
(C) 1372
(D) 1270

Q.14

If the volume and surface area of a sphere are numerically the same, then find the volume (in cm^3) of its semi – sphere. (correct to two places of decimals, use $\pi = 3.14$)

(A) 56.25
(B) 65.52
(C) 56.52
(D) 65.25

Q.15

The average age of 12 men is increased by 6 years when three of them, whose ages are 36, 40 and 44 years, are replaced by 3 persons. What is the average age of the 3 persons?

(A) 64 years
(B) 50 years
(C) 58 years
(D) 67 years

Q.16

If the cash difference between the selling price of an item at a gain of 3% and 6% is ₹9, which of the following options is the correct ratio between the selling prices?

(A) 103 : 106
(B) 101 : 105
(C) 101 : 103
(D) 103 : 100

Q.17

The average age of a group of 32 women is 60 years and that of another group of 40 men is 33 years. The average age of the two groups mixed together is:

(A) 46.5 years
(B) 48 years
(C) 45 years
(D) 42 years

Q.18

In a student council election, there were three contestants. All the voters participated in the voting and 10% of votes were declared invalid. The contestants got the votes in the ratio 8 : 7 : 5. If the second candidate in the list got 4725 votes, the total number of valid votes is:

(A) 11,500
(B) 10,000
(C) 13,500
(D) 9,000

Q.19

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 6% per year rate of interest for 4 years.

(A) 680
(B) 660
(C) 720
(D) 760

Q.20

Two vessels, A and B, contain spirit and water in the ratio 3 : 4 and 5 : 3, respectively. Find the ratio in which these two mixtures be mixed to obtain a new mixture in vessel C containing spirit and water in the ratio 4 : 3.

(A) 5 : 8
(B) 3 : 7
(C) 3 : 8
(D) 4 : 7

Reasoning Ability and Problem Solving 15 questions

Q.1

In a certain code language, 'MOVE' is coded as '3917' and 'EAST' is coded as '4692'. What is the code for 'E in the given code language?

(A) 7
(B) 2
(C) 9
(D) 4

Q.3

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All planets are rocks. All rocks are lakes. Conclusions: (I) All planets are lakes. (II) All rocks are planets.

(A) Neither conclusion I nor II follows.
(B) Only conclusion I follows.
(C) Only conclusion II follows.
(D) Both conclusions I and II follow.

Q.4

Seven friends, Kamala, Raveena, Amruta, Nisha, Gautami, Bhavana and Triveni, are sitting in a circular dining table facing the centre (but not necessarily in the same order). Raveena is sitting second to the right of Nisha. Gautami has Kamala and Triveni as immediate neighbours. Triveni is sitting third to the left of Nisha. Amruta is not an immediate neighbour of Nisha. Who is seated fourth to the left of Bhavana?

(A) Nisha
(B) Triveni
(C) Raveena
(D) Kamala

Q.5

In a certain code language, 'father makes tea' is coded as 'ib fx ey' and 'tea and coffee' is coded as 'kr ey wn'. How is 'tea' coded in the given language?

(A) kr
(B) fx
(C) ey
(D) wn

Q.6

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 8 + 2 – 1 × 1 ÷ 1 = ?

(A) 5
(B) 3
(C) 7
(D) 10

Q.7

M, N, O, P, Q and R are sitting around a circular table facing the centre (but not necessarily in the same order). N sits to the immediate right of R. M sits second to the right of N. P sits second to the right of M. O sits third to the right of P. Who sits to the immediate right of Q?

(A) O
(B) P
(C) R
(D) M

Q.9

PLRN is related to QMSO in a certain way based on the English alphabetical order. In the same way, LHNJ is related to MIOK. To which of the following is HDJF related, following the same logic?

(A) EIKG
(B) EIGK
(C) IEKG
(D) IEGK

Q.10

What will come in the place of the question mark (?) in the following equation, if '+' and '×' are interchanged and '÷' and '–' are interchanged? 15 ÷ 5 + 8 – 4 × 7 = ?

(A) 22
(B) 12
(C) 18
(D) 25

Q.11

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 24, 73; 27, 82

(A) 33, 101
(B) 33, 100
(C) 32, 100
(D) 32, 101

Q.12

519 is related to 781 following a certain logic. Following the same logic, 642 is related to 904. To which ofthe following is 485 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting / multiplying etc., to 13 canbe performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3is not allowed.)

(A) 756
(B) 721
(C) 794
(D) 747

Q.13

What should come in place of the question mark (?) in the given series based on the English alphabetical order? GGB, HHD ,IKH, JPN, ?

(A) PUV
(B) OUV
(C) OWV
(D) KWV

Q.16

FI29 is related to TW49 in a certain way. In the same way, LO38 is related to ZC58. To which of the following is RU65 related, following the same logic?

(A) FI85
(B) FE85
(C) EF58
(D) IF58

Q.18

What should come in place of the question mark (?) in the following series? 1,17, 33, 49, 65 ,?

(A) 80
(B) 79
(C) 81
(D) 77

Q.19

In a certain code language, 'GIFT' is coded as '5197' and 'FIST' is coded as '4157'. What is the code for 'S' in that language?

(A) 4
(B) 9
(C) 1
(D) 5

Q.20

In a certain code language, 'FLOAT' is coded as '14238' and 'BLOAT' is coded as '23849'. What is the code for 'F' in the given code language?

(A) 2
(B) 9
(C) 4
(D) 1

General Awareness 25 questions

Q.1

The sister of Harshavardhana Rajyashri was married to_____, the king of Maukhari dynasty of Kanyakubja.

(A) Sasanka
(B) Devagupta
(C) Grahavarman
(D) Isanavarman

Q.2

Sarhul is an agricultural festival of tribals, which is primarily celebrated in the Indian state of _____.

(A) Gujarat
(B) Uttarakhand
(C) Sikkim
(D) Jharkhand

Q.3

Which of the following states had benefitted by the Indira Gandhi Canal?

(A) Uttar Pradesh
(B) Gujarat
(C) Rajasthan
(D) Madhya Pradesh

Q.4

What is Cash Reserve Ratio?

(A) Percentage of deposits that a bank must keep as cash reserves with the central bank
(B) Percentage of deposits that a bank must keep as cash reserves with any bank
(C) Percentage of deposits that a central bank must keep as cash reserves with the government
(D) Percentage of deposits that a central bank must keep as cash reserves with the bank

Q.5

Which of the following mountain peaks is a part of the Himalayas?

(A) Kamet
(B) Taramati
(C) Kalsubai
(D) Dhupgarh

Q.6
What is the primary gas responsible for the leavening effect in bread when yeast ferments sugars?

(A) Carbon dioxide
(B) Helium
(C) Nitrogen
(D) Oxygen

Q.7
Which of the following is NOT related to goals of Fundamental Duties of Indian Citizens?

(A) Respect
(B) Cherish
(C) Courage
(D) Preserve

Q.8
Who among the following signed a pact with Lord Irwin on March 5, 1931?

(A) Subhas Chandra Bose
(B) Mahatma Gandhi
(C) B R Ambedkar
(D) Jawaharlal Nehru

Q.9
The digestive enzymes inside lysosomes are made by:

(A) mitochondria
(B) plastids themselves
(C) Golgi apparatus
(D) rough endoplasmic reticulum

Q.10
_____was well known for creating fusion art by adapting European theatrical techniques to Indian classical dance combined with Indian folk and tribal dance.

(A) Guru Bipin Singh
(B) Kelucharan Mohapatra
(C) Birju Maharaj
(D) Uday Shankar

Q.11
Which of the following is the style of temple architecture that is widely favoured in northern India?

(A) Vihar
(B) Nagar
(C) Madapa
(D) Dravid

Q.12
In the inaugural year of the Nehru Cup event for football in 1982, which team emerged as the winner of the first edition?

(A) Australia
(B) Poland
(C) Ghana
(D) Uruguay

Q.13
S Abdul Nazeer was appointed as the Governor of _____ in February 2023.

(A) Kerala
(B) Maharashtra
(C) Goa
(D) Andhra Pradesh

Q.14
Who was the Registrar General and Census Commissioner of India for the 2011 Indian Census?

(A) Devender Kumar Sikri
(B) Jayant Kumar Banthia
(C) Vivek Joshi
(D) Dr. C Chandramouli

Q.15
Which team won the Women's Asian Hockey Champions Trophy in 2023?

(A) South Korea
(B) Japan
(C) India
(D) Malaysia

Q.16
According to Census 2011, what is the sex ratio of India?

(A) 950
(B) 943
(C) 933
(D) 998

Q.17
'My Country My Life' is the autobiography of:

(A) Manmohan Singh
(B) Lal Krishna Advani
(C) Natwar Singh
(D) Khushwant Singh

Q.18
In which of the following systems was the charge of collecting the revenue and paying it to the Company given to the village headman, rather than the zamindar?

(A) Ryotwari
(B) Mahalwari
(C) Zamindari
(D) Iqtadari

Q.19
Helium gas can be converted to the liquid form by the _____.

(A) London forces of attraction
(B) dipole-dipole forces of attraction
(C) hydrogen bonding interaction
(D) dipole-induced dipole forces of attraction

Q.20
Which Tamil superstar founded the Tamilaga Vettri Kazhagam (TVK) political party in Tamil Nadu and Puducherry on 2 February 2024 with the slogan 'Pirappokkum Ellaa Uyirkkum' (All are equal by birth)?

(A) Vijay
(B) Kamal Haasan
(C) Dhanush
(D) Rajinikanth

Q.21
A river is capable of forming which of the following landscapes when the river is in its middle course?

(A) V shaped valley
(B) A delta
(C) Waterfalls
(D) A meander

Q.22
Kalamandalam Krishnan Nair was a notable figure in which of the following classical dance forms of India?

(A) Bharatnatyam
(B) Kuchipudi
(C) Mohiniyattam
(D) Kathakali

Q.23
Pandit Arvind Thatte belongs to which of the following gharanas?

(A) Benaras gharana
(B) Mewati gharana
(C) Agra gharana
(D) Indore gharana

Q.24
Who was the famous Venetian traveller of the 13th century who visited Kerala?

(A) Barthema
(B) Abdur Razzaq
(C) Nicolo Conti
(D) Marco Polo

Q.25
Who among the following is the ex-officio Chairperson of the Rajya Sabha as on 10 October 2023?

(A) Jagdeep Dhankar
(B) Prakash Javdekar
(C) Virendra Kumar
(D) Piyush Goyal

English Language and Comprehension 25 questions

Q.1

Select the most appropriate option to fill in the blank School classrooms have _____ very little over the last century.

(A) changed
(B) freed
(C) startled
(D) unconcerned

Q.2

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. He is waiting / at airport to receive / Mr. Smith, who is / coming from South Africa.

(A) at airport to receive
(B) He is waiting
(C) coming from South Africa.
(D) Mr. Smith, who is

Q.3

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. He cheated in the [A] / examination, making his [B] / already-falling academic [C] / credentials worst. [D]

(A) D
(B) B
(C) A
(D) C

Q.4

Select the most appropriate ANTONYM of the given word. Entice

(A) Allure
(B) Invite
(C) Repulse
(D) Tangle

Q.5

Select the most appropriate meaning of the underlined idiom. He “gnashed his teeth” at his son's actions.

(A) expressed sadness
(B) expressed joy
(C) expressed disappointment
(D) expressed rage

Q.6

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. The new almirah is “big” than this one.

(A) bigger
(B) biggest
(C) more big
(D) more bigger

Q.7

Select the correctly spelt word to fill in the blank. We are entering a point where technology will change at _____ speeds.

(A) extreme
(B) inferier
(C) modarate
(D) suprime

Q.8

Select the most appropriate meaning of the given idiom. Run errands

(A) To go out to buy or do something
(B) To participate in a race through unfair means
(C) To finish running before the end point
(D) To do something incorrectly or erroneous in haste

Q.9

Identify the ODD word from the given options.

(A) Carom
(B) Table tennis
(C) Billiards
(D) Golf

Q.10

Select the most appropriate option to fill in the blank. Leisure activities can improve an individual's physical and mental health by _____ stress, and offering a pleasurable experience.

(A) reconstructing
(B) irritating
(C) upsetting
(D) relieving

Q.11

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. My friend, / Priya looks more cheerful today / as she celebrates her birthday.

(A) No error
(B) as she celebrates her birthday
(C) Priya looks more cheerful today
(D) My friend,

Q.12

Select the most appropriate ANTONYM of the given word. Conscious

(A) Sentimental
(B) Sensible
(C) Unaware
(D) Secure

Q.13

The following sentence has been split into four segments. Identify the segment that contains anerror in phrasal verb. Tina dropped behind of / college and / went straight / into a good job.

(A) into a good job
(B) Tina dropped behind of
(C) college and
(D) went straight

Q.14

Select the most appropriate synonym of the word given in brackets to fill in the blank. The rivals in the institution _____ (fabricated) evidence and manipulated the whole dataset.

(A) fluctuated
(B) wrecked
(C) destroyed
(D) forged

Q.15

Select the option that rectifies the spelling of the underlined word in the given sentence. The concert was "canceeled" due to rain.

(A) Cancealed
(B) Cancelled
(C) Cencelled
(D) Cancaeled

Q.16

Select the INCORRECTLY spelt word.

(A) Intransigence
(B) Spouse
(C) Endengered
(D) Unambiguous

Q.17

Select the most appropriate degree of comparison to fill in the blank. Many ornithologists believe that the northern mocking birds are _____ wild birds in the jungle.

(A) noisy
(B) the noisiest
(C) the noisy
(D) noisier

Q.18

Select the most appropriate option to fill in the blanks. Our _____ conducted a _____ to get everyone's opinion.

(A) principal; poll
(B) principal; pole
(C) principle; poll
(D) principle; pole

Q.19

Select the option that will improve the underlined part of the given sentence. Arun asks Smita to copy the contents of the document “word by word”.

(A) word with word
(B) word par word
(C) word for word
(D) word to word

Q.20

Select the most appropriate ANTONYM to substitute the underlined word in the given sentence. In what ways do you find technological advancements “beneficial” to your daily life?

(A) unfathomable
(B) unwarranted
(C) unfavourable
(D) unapproachable

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Dolphins 1_____ regarded as the friendliest creatures in the sea, and stories of them helping drowning sailors have been common since Roman times. The more we 2_____ about dolphins, the more we realise that their society is more complex 3_____ people previously imagined. They look 4_____ other dolphins when they are ill, care for pregnant 5_____ and protect the weakest in the community, as we do.

Select the most appropriate option to fill in blank number 1.

(A) has
(B) was
(C) is
(D) are

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Dolphins 1_____ regarded as the friendliest creatures in the sea, and stories of them helping drowning sailors have been common since Roman times. The more we 2_____ about dolphins, the more we realise that their society is more complex 3_____ people previously imagined. They look 4_____ other dolphins when they are ill, care for pregnant 5_____ and protect the weakest in the community, as we do.

Select the most appropriate option to fill in blank number 2.

(A) learn
(B) learned
(C) learning
(D) learns

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Dolphins 1_____ regarded as the friendliest creatures in the sea, and stories of them helping drowning sailors have been common since Roman times. The more we 2_____ about dolphins, the more we realise that their society is more complex 3_____ people previously imagined. They look 4_____ other dolphins when they are ill, care for pregnant 5_____ and protect the weakest in the community, as we do.

Select the most appropriate option to fill in blank number 3.

(A) to
(B) the
(C) for
(D) than

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Dolphins 1_____ regarded as the friendliest creatures in the sea, and stories of them helping drowning sailors have been common since Roman times. The more we 2_____ about dolphins, the more we realise that their society is more complex 3_____ people previously imagined. They look 4_____ other dolphins when they are ill, care for pregnant 5_____ and protect the weakest in the community, as we do.

Select the most appropriate option to fill in blank number 4.

(A) on
(B) into
(C) after
(D) up

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Dolphins 1_____ regarded as the friendliest creatures in the sea, and stories of them helping drowning sailors have been common since Roman times. The more we 2_____ about dolphins, the more we realise that their society is more complex 3_____ people previously imagined. They look 4_____ other dolphins when they are ill, care for pregnant 5_____ and protect the weakest in the community, as we do.

Select the most appropriate option to fill in blank number 5.

(A) brothers
(B) mothers
(C) fathers
(D) uncles

Answer key

Numerical and Mathematical Ability

Q.1 **(B)**	Q.2 **(C)**	Q.4 **(C)**	Q.5 **(C)**	Q.6 **(B)**	Q.7 **(D)**
Q.8 **(A)**	Q.9 **(A)**	Q.10 **(D)**	Q.12 **(C)**	Q.14 **(C)**	Q.15 **(A)**
Q.16 **(A)**	Q.17 **(C)**	Q.18 **(C)**	Q.19 **(C)**	Q.20 **(C)**	

Reasoning Ability and Problem Solving

Q.1 **(C)**	Q.3 **(B)**	Q.4 **(B)**	Q.5 **(C)**	Q.6 **(C)**	Q.7 **(B)**
Q.9 **(C)**	Q.10 **(B)**	Q.11 **(B)**	Q.12 **(D)**	Q.13 **(D)**	Q.16 **(A)**
Q.18 **(C)**	Q.19 **(A)**	Q.20 **(D)**			

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(C)**	Q.4 **(A)**	Q.5 **(A)**	Q.6 **(A)**
Q.7 **(C)**	Q.8 **(B)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(B)**	Q.12 **(D)**
Q.13 **(D)**	Q.14 **(D)**	Q.15 **(C)**	Q.16 **(B)**	Q.17 **(B)**	Q.18 **(B)**
Q.19 **(A)**	Q.20 **(A)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(D)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(A)**
Q.7 **(A)**	Q.8 **(A)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(A)**	Q.12 **(C)**
Q.13 **(B)**	Q.14 **(D)**	Q.15 **(B)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(A)**
Q.19 **(C)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(C)**
Q.25 **(B)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (B) 1350

SI = 2,500 × 9 × 6/100 = ₹1,350.

Q.2 · Answer (C) ₹2,56,675

The cost price is ₹1,54,005 ÷ 0.75 = ₹2,05,340. For a 25% profit he must sell the laptops for ₹2,05,340 × 1.25 = ₹2,56,675.

Q.4 · Answer (C) 1 : 4

Selling at cost price, all the gain comes from the water: to gain 25% he must sell 125 litres for the price of 100 litres of milk, so water : milk = 25 : 100 = 1 : 4.

Q.5 · Answer (C) 480

If ₹x is lent at 25%: 0.25x + 0.20(2,380 – x) = 500, so 0.05x = 500 – 476 = 24 and x = 480.

Q.6 · Answer (B) 12

84 = $2^2 \times 3 \times 7$ and 108 = $2^2 \times 3^3$; the common factors give $2^2 \times 3 = 12$.

Q.7 · Answer (D) 224 cm²

The circle's area is (22/7) × 14^2 = 616 cm². The largest inscribed square has the diameter, 28 cm, as its diagonal, so its area is $28^2/2$ = 392 cm². 616 – 392 = 224 cm².

Q.8 · Answer (A) 3.6

Downstream speed = 15 + 3 = 18 km/h; in 12 minutes (1/5 hour) the boat covers 18/5 = 3.6 km.

Q.9 · Answer (A) 30

With each part d km: d/3 + d/4 + d/8 = 17d/24 hours = 425/60 hours, so d = 10 and the total distance is 3 × 10 = 30 km.

Q.10 · Answer (D) ₹3,840

Aman paid ₹18,000 × 0.85 × 0.90 = ₹13,770, plus ₹830 for transport: ₹14,600 in all. Profit = ₹18,440 – ₹14,600 = ₹3,840.

Q.12 · Answer (C) 1372

2πr = 264 gives the bowl's radius as 42 cm. Half its volume is ½ × (2/3)π × 42^3 = 24,696π cm³, and each glass holds (2/3)π × 3^3 = 18π cm³, so 24,696π ÷ 18π = 1372 people are served.

Q.14 · Answer (C) 56.52

(4/3)πr^3 = 4πr^2 gives r = 3 cm. The hemisphere's volume is (2/3)π × 3^3 = 18π = 18 × 3.14 = 56.52 cm³.

Q.15 · Answer (A) 64 years

The total rose by 12 × 6 = 72 years. The three men replaced total 36 + 40 + 44 = 120 years, so the three newcomers total 192: an average of 64 years.

Q.16 · Answer (A) 103 : 106

At gains of 3% and 6% the selling prices are 1.03 and 1.06 times the cost price, so their ratio is 103 : 106 whatever the cost (here ₹300, since 3% of it is ₹9).

Q.17 · Answer (C) 45 years

(32 × 60 + 40 × 33)/72 = (1,920 + 1,320)/72 = 3,240/72 = 45 years.

Q.18 · Answer (C) 13,500

The second candidate's 7 parts are 4725 votes, so one part is 675. The valid votes are 8 + 7 + 5 = 20 parts: 20 × 675 = 13,500.

Q.19 · Answer (C) 720

SI = 3,000 × 6 × 4/100 = ₹720.

Q.20 · Answer (C) 3 : 8

Spirit is 3/7 of A, 5/8 of B and must be 4/7 of the mixture. By alligation A : B = (5/8 − 4/7) : (4/7 − 3/7) = 3/56 : 8/56 = 3 : 8.

Reasoning Ability and Problem Solving

Q.1 · Answer (C) 9

E is the only letter in both MOVE and EAST, and 9 the only digit in both 3917 and 4692, so E = 9.

Q.3 · Answer (B) Only conclusion I follows.

All planets are rocks and all rocks are lakes, so all planets are lakes (I). Nothing says every rock is a planet, so II does not follow.

Q.4 · Answer (B) Triveni

Triveni is third to Nisha's left and Raveena second to her right. Gautami and Kamala cannot both fit in the one seat between Triveni and Raveena, so going left from Nisha come Kamala, Gautami, Triveni; Amruta cannot sit beside Nisha, so she takes the seat after Triveni and Bhavana is on Nisha's right. Four seats to Bhavana's left: Nisha, Kamala, Gautami, Triveni.

Q.5 · Answer (C) ey

'tea' is the only word in both sentences and 'ey' the only code in both, so tea = ey.

Q.6 · Answer (C) 7

After interchanging the signs: 8 − 2 + 1 ÷ 1 × 1 = 8 − 2 + 1 = 7.

Q.7 · Answer (B) P

Going round to the right from R: N, O, M, Q, P. M is second to N's right and P second to M's right; O, third to P's right, takes the seat between N and M, which leaves Q between M and P. P is to Q's immediate right.

Q.9 · Answer (C) IEKG

Each letter moves one place forward: P → Q, L → M, R → S, N → O. So HDJF → IEKG.

Q.10 · Answer (B) 12

After interchanging the signs: 15 − 5 × 8 ÷ 4 + 7 = 15 − 10 + 7 = 12.

Q.11 · Answer (B) 33, 100

The second number is three times the first plus 1: 24 × 3 + 1 = 73 and 27 × 3 + 1 = 82. Likewise 33 × 3 + 1 = 100.

Q.12 · Answer (D) 747

Each number is related to itself plus 262: 519 + 262 = 781 and 642 + 262 = 904. So 485 + 262 = 747.

Q.13 · Answer (D) KWV

First letters run G, H, I, J, so next is K; second letters move +1, +3, +5 (G, H, K, P), so next is P + 7 = W; third letters move +2, +4, +6 (B, D, H, N), so next is N + 8 = V: KWV.

Q.16 · Answer (A) FI85

Each letter moves 14 places forward (wrapping round after Z) and the number rises by 20: F → T, I → W, 29 → 49 and L → Z, O → C, 38 → 58. So R → F, U → I and 65 → 85: FI85.

Q.18 · Answer (C) 81

Each term is 16 more than the one before: 65 + 16 = 81.

Q.19 · Answer (A) 4

GIFT and FIST share I, F and T, and 5197 and 4157 share 1, 5 and 7. The letter left in FIST is S and the digit left in its code is 4, so S = 4.

Q.20 · Answer (D) 1

FLOAT and BLOAT share L, O, A and T, and 14238 and 23849 share 2, 3, 4 and 8. The letter left in FLOAT is F and the digit left in its code is 1, so F = 1.

General Awareness

Q.1 · Answer (C) Grahavarman

Rajyashri, Harshavardhana's sister, was married to Grahavarman, the Maukhari king of Kanauj (Kanyakubja).

Q.2 · Answer (D) Jharkhand

Sarhul, a spring festival in which the blossoms of the sal tree are worshipped, is celebrated mainly by the tribes of Jharkhand.

Q.3 · Answer (C) Rajasthan

The Indira Gandhi Canal, which starts at the Harike Barrage in Punjab, carries water into the Thar Desert of western Rajasthan.

Q.4 · Answer (A) Percentage of deposits that a bank must keep as cash reserves with the central bank

The Cash Reserve Ratio is the share of its deposits that a bank must keep as cash with the central bank (in India, the Reserve Bank of India).

Q.5 · Answer (A) Kamet

Kamet (7,756 m) is in the Garhwal Himalaya of Uttarakhand. Kalsubai and Taramati are in the Western Ghats of Maharashtra, and Dhupgarh is the highest point of the Satpura range.

Q.6 · Answer (A) Carbon dioxide

Yeast ferments sugars into alcohol and carbon dioxide; the bubbles of carbon dioxide make the dough rise.

Q.7 · Answer (C) Courage

Article 51A asks citizens to abide by the Constitution and respect its ideals, the National Flag and the National Anthem, to cherish the ideals of the freedom struggle and to value and preserve the country's heritage; 'courage' is not among the duties.

Q.8 · Answer (B) Mahatma Gandhi

The Gandhi-Irwin Pact, signed on 5 March 1931, ended the first phase of the Civil Disobedience Movement, and Gandhi went on to attend the Second Round Table Conference.

Q.9 · Answer (D) rough endoplasmic reticulum

The digestive enzymes in lysosomes are proteins made by the rough endoplasmic reticulum.

Q.10 · Answer (D) Uday Shankar

Uday Shankar created a fusion style of dance, adapting European theatrical techniques to Indian classical dance with elements of folk and tribal dance.

Q.11 · Answer (B) Nagar

The Nagara style, with its curving shikhara, prevails in northern India; the Dravida style is the south's.

Q.12 · Answer (D) Uruguay

Uruguay won the first Nehru Cup, held in Kolkata in 1982, beating China in the final.

Q.13 · Answer (D) Andhra Pradesh

S. Abdul Nazeer, a retired Supreme Court judge, was appointed Governor of Andhra Pradesh in February 2023.

Q.14 · Answer (D) Dr. C Chandramouli

Dr C. Chandramouli was the Registrar General and Census Commissioner of India for the 2011 Census.

Q.15 · Answer (C) India

India won the 2023 Women's Asian Champions Trophy at Ranchi, beating Japan 4-0 in the final.

Q.16 · Answer (B) 943

Census 2011 recorded 943 females per 1,000 males in India.

Q.17 · Answer (B) Lal Krishna Advani

'My Country My Life' (2008) is the autobiography of L. K. Advani.

Q.18 · Answer (B) Mahalwari
Under the Mahalwari system, introduced in the North-Western Provinces in 1822, the village headman collected the revenue of the whole village (mahal) and paid it to the Company.

Q.19 · Answer (A) London forces of attraction
Helium atoms are non-polar, so the only forces between them are weak London (dispersion) forces; these are what let helium be liquefied.

Q.20 · Answer (A) Vijay
The actor Vijay launched Tamilaga Vettri Kazhagam on 2 February 2024. Kamal Haasan's party is Makkal Needhi Maiam, founded in 2018.

Q.21 · Answer (D) A meander
In its middle course a river flows over gentler slopes and swings from side to side in meanders. V-shaped valleys and waterfalls belong to the upper course, deltas to the mouth.

Q.22 · Answer (D) Kathakali
Kalamandalam Krishnan Nair was one of the great Kathakali artists of Kerala.

Q.23 · Answer (B) Mewati gharana
The harmonium player Arvind Thatte, a disciple of Pandit Jasraj since 1981, belongs to Jasraj's Mewati gharana.

Q.24 · Answer (D) Marco Polo
Marco Polo of Venice visited the Malabar coast of Kerala late in the 13th century. Nicolo Conti was a 15th-century Venetian, Varthema (Barthema) came from Bologna in the 16th century, and Abdur Razzaq was a 15th-century Persian envoy.

Q.25 · Answer (A) Jagdeep Dhankar
The Vice-President is the ex-officio Chairman of the Rajya Sabha; Jagdeep Dhankhar has held the office since August 2022.

English Language and Comprehension

Q.1 · Answer (A) changed
Classrooms have 'changed' very little: the sentence is about how little they have altered in a century.

Q.2 · Answer (A) at airport to receive
A particular airport takes the article: 'He is waiting at the airport'.

Q.3 · Answer (A) D
'Making his credentials worse' needs the comparative: credentials that were already falling became 'worse', not 'worst'.

Q.4 · Answer (C) Repulse
To 'entice' is to attract or tempt; its opposite is to 'repulse', to drive away.

Q.5 · Answer (D) expressed rage
To 'gnash one's teeth' is to show anger or rage.

Q.6 · Answer (A) bigger
A comparison with 'than' takes the comparative: 'bigger than this one'.

Q.7 · Answer (A) extreme
'Extreme' is spelt correctly; the others should be 'inferior', 'moderate' and 'supreme'.

Q.8 · Answer (A) To go out to buy or do something
To 'run errands' is to make short trips to buy or do things, often for someone else.

Q.9 · Answer (D) Golf
Carom, table tennis and billiards are indoor games played on a table or board; golf is played outdoors on a course.

Q.10 · Answer (D) relieving
Leisure improves health by 'relieving', that is easing, stress.

Q.11 · Answer (A) No error
The sentence is correct as it stands: 'Priya looks' agrees, 'more cheerful today' compares her with her usual mood, and 'as she celebrates' gives the reason.

Q.12 · Answer (C) Unaware
'Conscious' means aware; its opposite is 'unaware'.

Q.13 · Answer (B) Tina dropped behind of
To leave college before finishing is to 'drop out of' it: 'Tina dropped out of college'.

Q.14 · Answer (D) forged
'Fabricated' evidence is 'forged': made up to deceive.

Q.15 · Answer (B) Cancelled
The correct spelling is 'cancelled'.

Q.16 · Answer (C) Endengered
The correct spelling is 'endangered'.

Q.17 · Answer (B) the noisiest
A comparison with all the wild birds in the jungle takes the superlative: 'the noisiest'.

Q.18 · Answer (A) principal; poll
The head of a school is the 'principal' (a 'principle' is a rule), and a survey of opinion is a 'poll' (a 'pole' is a post).

Q.19 · Answer (C) word for word
To copy something exactly, in the same words, is to copy it 'word for word'.

Q.20 · Answer (C) unfavourable
'Beneficial' means helpful or advantageous; of the options only 'unfavourable' is opposite in meaning.

Q.21 · Answer (D) are
The plural subject takes 'are': 'Dolphins are regarded as the friendliest creatures'.

Q.22 · Answer (A) learn
'The more we learn about dolphins, the more we realise': the present tense, like 'realise'.

Q.23 · Answer (D) than
'More complex' is followed by 'than'.

Q.24 · Answer (C) after
To 'look after' is to take care of: dolphins look after others when they are ill.

Q.25 · Answer (B) mothers
Dolphins care for pregnant 'mothers'; the other options are all male.

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